



Data Article

Data health risk assessment of nitrate contamination in groundwater of rural region in the Yerraguntla Mandal, South India

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ARTICLE INFO

Article history:

Received 4 February 2020

Revised 24 February 2020

Accepted 25 February 2020

Available online 11 March 2020

Keywords:

Nitrates

Health risk

Groundwater

Yerraguntla

ABSTRACT

The main objective of the present study was to determine the distribution levels of nitrate contamination in groundwater and its associated impact on human health risk in the Yerraguntla mandal, South India. For this 40 ground water samples were collected randomly during April 2018. Nitrate concentration in groundwater samples ranged from 2.50 to 760.12 mg/L, with a mean value of 86.13 mg/L. Most of the groundwater samples are exceeding the permissible limits of nitrate (45 mg/l). Hence health risk assessment of nitrate has been carried out. Hazard quotient (HQ) values for infants, children, male and female ranges from 0.05 to 14.25; 0.06 to 18.53; 0.04 to 12.18 and 0.05 to 14.62, respectively. The finding of data showed that HQ value was more than 1 in 42.5% of samples in groups of infants, children and female, and 35% of samples in group of male.

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Specification table

Subject area	Yerraguntla
More specific subject area	Nitrate in groundwater
Type of data	Tables and figures
How data was acquired	40 Groundwater Samples collected in different bore wells at Yerraguntla Mandal, Y.S.R district, A.P with help of location map. Groundwater samples analyzed using a UV-visible spectro-Photometer according to standard methods for examination of groundwater
Data format	Raw and analyzed
Parameters for data collection	Groundwater samples were tightly packed in 2 L. Polyethylene plastic bottles, stored in Geo chemical lab at room Temperature and determined for nitrate concentration.
Description for data collection	Determine the concentration levels of nitrate
Data source Location	Yerraguntla Mandal, Y.S.R Kadapa District
Data accessibility	Data is included in this Article

Value of the data

- The data of present paper revealed that the nitrate concentration in 45% of groundwater samples exceeding the maximum permissible limits (45 mg/L) according to world health organization (WHO) guidelines.
- The findings of this study reveal that all the four groups i.e. infants, children, female and male are exposed to nitrate risk ($HQ>1$).
- High nitrate concentration in drinking water causes health risks on human body like methemoglobinemia in infants and stomach cancer in children.
- The findings of this study reveal the all the four groups i.e. infants, children, female and male are exposed to nitrate risk ($HQ>1$).
- Health risk assessment revealed that male, are less prone to risk than female. Hence proper precautionary measures have to be taken to control health risk in this area.

1. Data description

1.1. Study area

Yerraguntla is a limestone area located within the Kadapa district of the Andhra Pradesh. The study area lies between Latitudes N 14°35′ 17.2″–N 14°44′ 38.4″ and Longitudes E 78°27′38.4″–E 78°34′39.3″ (Fig. 1). In this area mainly Arenaceous consisting of conglomerate quartzite, Quartzite with shale formation of dolomitic lime stones are present. Groundwater quality may vary depending upon variations in geological formations. The total annual rainfall in the study area is 730.7 mm [1,2].

1.2. Analytical data

Spatial distribution map of the nitrate shown in Fig. 2; revealed that most of the higher nitrate concentration fall eastern part of the study area. The concentration of nitrate and HQ values for infants, children, female, male shown in Table 1. Table 2 show that the ranges of HQ for four groups. In the present study, 42.5% of the groundwater samples exceeding 1($HQ>1$) is referred as adverse non-carcinogenic risk for human health, for infants, children, female and 35% of samplers exceeding 1($HQ>1$) for male as shown in Fig. 3.

Table 1

Nitrate concentration and hazard quotient for the four groups of water consumers.

S. No	Name of the villages	Latitude	Longitude	Nitrate concentration (mg/L)	HQ Values			
					Infants	Children	Male	Female
1	Thippaluru	N 14°36' 46.3''	E 78°34' 39.3''	69.2	1.30	1.69	1.11	1.33
2	Sri Sai Charan School	N 14°37' 04.6''	E78°34' 38.4''	72.2	1.35	1.76	1.16	1.39
3	Sunkesula	N 14° 36' 19.7''	E78°34' 1.0''	65.23	1.22	1.59	1.05	1.25
4	Koduru	N 14°36' 2.7''	E78°32' 20.6''	65.34	1.23	1.59	1.05	1.26
5	Peddapadu	N 14°35' 17.2''	E 78°31' 28.1''	160.23	3.00	3.91	2.57	3.08
6	Tummalapalli	N 14°36' 23.3''	E 78°30' 35.6''	140.36	2.63	3.42	2.25	2.70
7	Tummalapalli	N 14°36' 26.2''	E 78°30' 30.9''	290.24	5.44	7.07	4.65	5.58
8	Elumkuripalli	N 14°36' 38.3''	E 78°30' 32.9''	250.24	4.69	6.10	4.01	4.81
9	Elumkuripalli	N 14°36' 1.5''	E 78°29' 30.7''	28	0.53	0.68	0.45	0.54
10	Tummalapalli	N 14°35' 51.7''	E 78°29' 5.1''	26	0.49	0.63	0.42	0.50
11	Valasapalli	N 14°37' 32.2''	E 78°31' 0''	36	0.68	0.88	0.58	0.69
12	Kadirivaripalle	N 14°37' 54.4''	E 78°30' 40.9''	140.24	2.63	3.42	2.25	2.70
13	Kadirivaripalle	N 14°37' 55.9''	E 78°30' 37.0''	36	0.68	0.88	0.58	0.69
14	Chirrajupalle	N 14°37' 37.9''	E 78°27' 53.2''	3.42	0.06	0.08	0.05	0.07
15	Chirrajupalle	N 14°37' 58.01''	E 78°27' 59.2''	9	0.17	0.22	0.14	0.17
16	Chilamkur	N 14°38' 51.3''	E 78°28' 6.4''	2.5	0.05	0.06	0.04	0.05
17	Chilamkur	N 14°38' 58.3''	E 78°27' 38.4''	6.3	0.12	0.15	0.10	0.12
18	Chilamkur	N 14°38' 58.3''	E 78°27' 38.4''	4.21	0.08	0.10	0.07	0.08
19	Niduzuvi	N 14°38' 52.0''	E 78°28' 6.2''	9.24	0.17	0.23	0.15	0.18
20	Chilamkur	N 14°38' 40.6''	E 78°30' 6.9''	8.2	0.15	0.20	0.13	0.16
21	Kalamala	N 14°41' 50.1''	E 78°28' 32.9''	10.12	0.19	0.25	0.16	0.19
22	Kalamala	N 14°41' 50.0''	E 78°28' 32.7''	10.24	0.19	0.25	0.16	0.20
23	Krishna Nagar(N C)	N 14°42' 32.3''	E 78°28' 27.7''	11.2	0.21	0.27	0.18	0.22
24	Mekalabalayapalem	N 14°42' 39.5''	E 78°28' 30.3''	17.24	0.32	0.42	0.28	0.33
25	Chinnadandluru	N 14°44' 38.4''	E 78°28' 52.2''	15.24	0.29	0.37	0.24	0.29
26	Chinnadandluru	N 14°44' 22.8''	E 78°29' 46.6''	9.24	0.17	0.23	0.15	0.18
27	Chinnadandluru	N 14°44' 22.8''	E 78°29' 46.9''	10.25	0.19	0.25	0.16	0.20
28	Illuru	N 14°44' 21.1''	E 78°30' 43.5''	11.24	0.21	0.27	0.18	0.22
29	Mallepadu	N 14°42' 34.3''	E 78°30' 23.9''	32.12	0.60	0.78	0.51	0.62
30	Mallepadu	N 14°42' 27.3''	E 78°30' 53.1''	36.26	0.68	0.88	0.58	0.70
31	Venkatapuram	N 14°42' 32.3''	E 78°31' 52.8''	110.24	2.07	2.69	1.77	2.12
32	Potladurti	N 14°42' 27.6''	E 78°32' 45.6''	560.24	10.50	13.66	8.98	10.77
33	Potladurti	N 14°42' 28.5''	E 78°32' 53.5''	26	0.49	0.63	0.42	0.50
34	Bismillakhaderpalli	N 14°41' 42.8''	E 78°33' 44.6''	40	0.75	0.98	0.64	0.77
35	Hanumangutta	N 14°40' 55.7''	E 78°34' 22.0''	760.12	14.25	18.53	12.18	14.62
36	Yerraguntla	N 14°38' 35.0''	E 78°32' 37.5''	120.24	2.25	2.93	1.93	2.31
37	Yerraguntla	N 14°38' 34.2''	E 78°32' 35.9''	58	1.09	1.41	0.93	1.12
38	Yerraguntla	N 14°38' 18.8''	E 78°32' 19.1''	56	1.05	1.37	0.90	1.08
39	Yerraguntla	N 14°37' 59.8''	E 78°32' 42.8''	62	1.16	1.51	0.99	1.19
40	Thermal waste water	N 14°42' 26.3''	E 78°28' 27.7''	67	1.26	1.63	1.07	1.29

Table 2

Classification of HQ ranges for Infants, children, male, female.

Human	Range of HQ	Health risk	Sample numbers	No of samples	% of samples
Infants	<1	No risk	9 to 11, 13 to 30, 33, 34	23	57.5
	>1	High risk	1 to 8, 12, 31, 32, 35 to 40	17	42.5
Children	<1	No risk	9 to 11, 13 to 30, 33, 34	23	57.5
	>1	High risk	1 to 8, 12, 31, 32, 35 to 40	17	42.5
Male	<1	No risk	9 to 11, 13 to 30, 33, 34, 37, 38, 39	26	65
	>1	High risk	1 to 8, 12, 31, 32, 35, 36, 40	14	35
Female	<1	No risk	9 to 11, 13 to 30, 33, 34	23	57.5
	>1	High risk	1 to 8, 12, 31, 32, 35 to 40	17	42.5

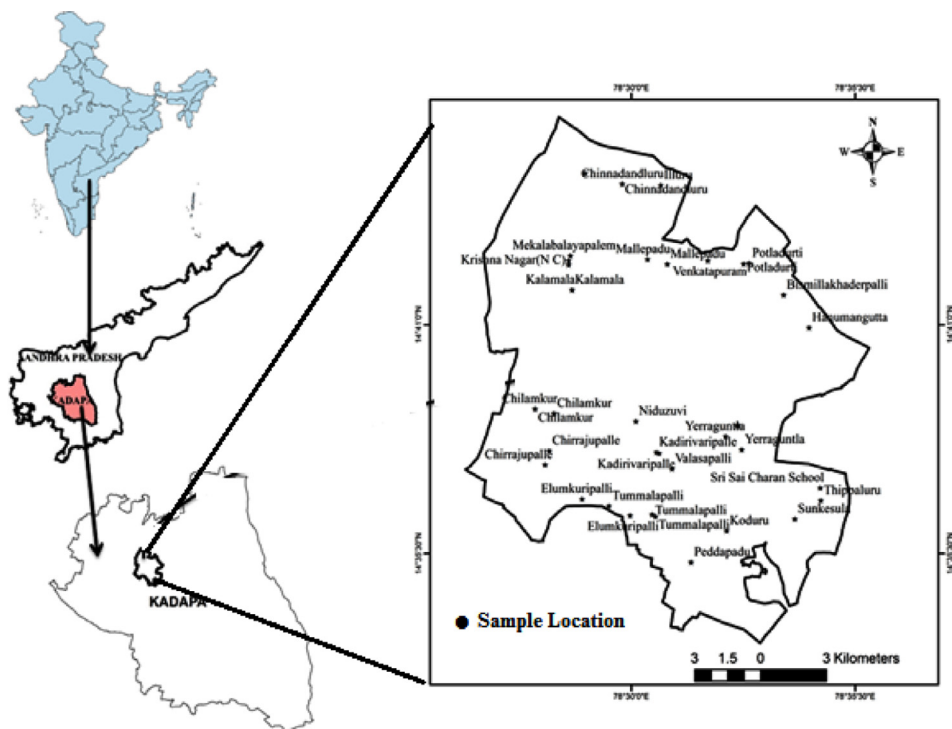


Fig. 1. Location map of the study area (Yerraguntla).

2. Experimental design, materials, and methods

The Samples were collected from 40 points in the different villages of Yerraguntla Mandal (Fig. 1). The samples were collected systematically during the month of April from bore/ hand pumps of study area. The samples were collected in acid washed 1 liter polyethylene bottles to prevent unpredictable changes in characteristic as per standard procedures [3]. Each of the groundwater samples was analyzed for nitrate. The nitrate concentration was measured using Spectro- photometer accordance standard methods for examination of groundwater [4–7].

2.1. Human health risk assessment

In recent decades, groundwater quality has continued to deteriorate due to the various pollution sources such as chemicals and fertilizers [8]. And due to this there is an increased awareness of health risk on humans. Health risk assessment plays a crucial role in this new era. Health assessment insures and evaluates adverse human health effect on infants, children, female and male [9,10]. This study signifies the health risk assessment associated with NO_3^- in drinking water and around Yerraguntla Mandal. The chronic daily intake (CDI) and hazard quotient (HQ)

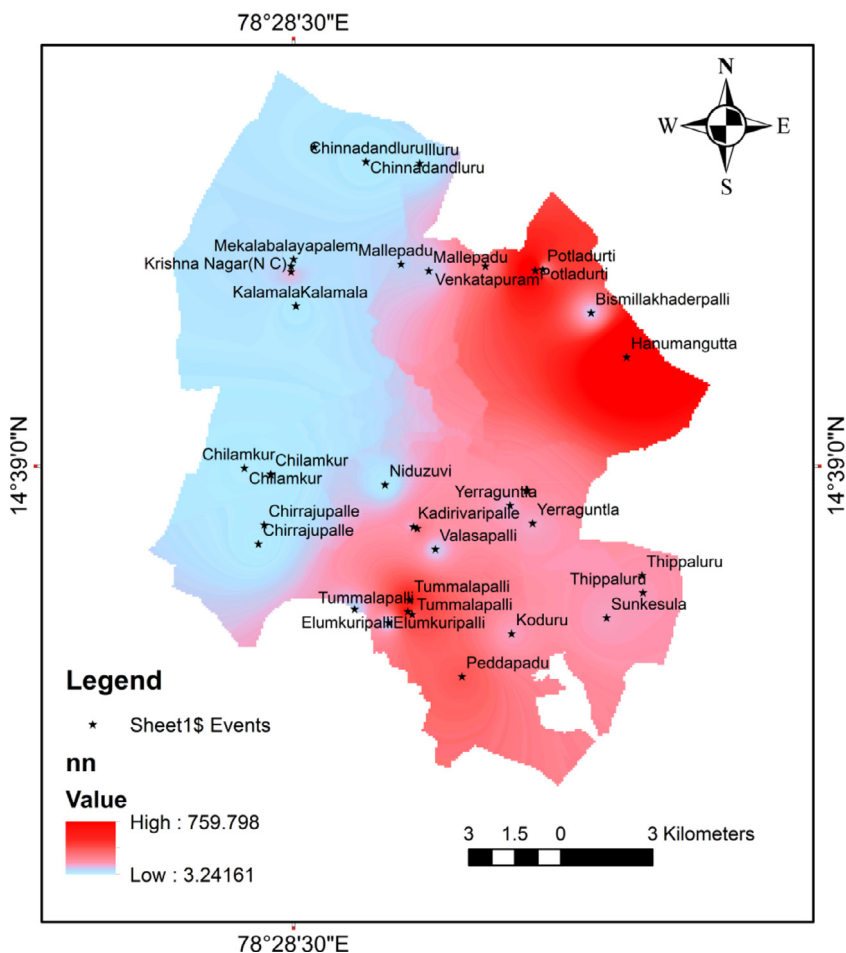


Fig. 2. Spatial distribution map of nitrate concentration.

were calculated by following formulas:

$$CDI = \frac{CPW \times IR \times ED \times EF}{ABW \times AET}$$

$$HQ = \frac{CDI}{RfD}$$

Where CDI is the chronic daily intake (mg/kg/day), CPW indicate nitrate contamination concentration in groundwater (mg/L), Ingestion rate (IR) 2L/day for male and female, 0.78 L/day for children, 0.3 L/day infants [11]. ED indicates exposure duration Years 40, 40, 12 & <1 for male, female, children, infants respectively [12]. EF indicates exposure frequency is assigned, 365 days/year for male, female, children and Infants, ABW denotes average body weight of human body in kg 78 (male), 65 (female), 20 (Children), 10 (infants), AET denotes average exposure time Days, 14,600 for male and female, 4380 children and 365 infants. RfD indicates that reference of NO_3^- (1.6 mg/kg/d) were obtained from the database of Integrated Risk Information System (IRIS), and USEPA (2012) [13]. Hazard quotient value exceeding 1 is referred as adverse non-carcinogenic risk for human health, while HQ values less than 1 indicates acceptable limit of non-carcinogenic risk.

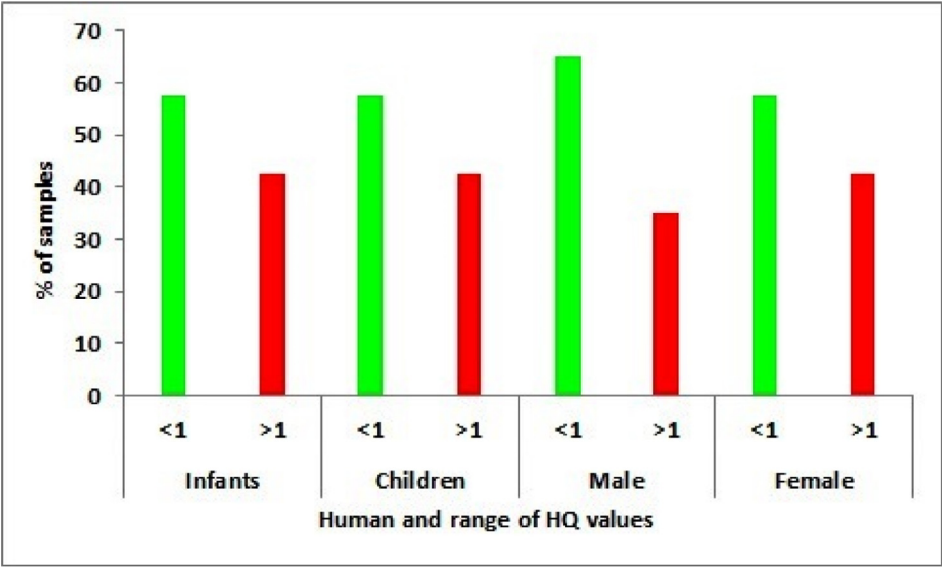


Fig. 3. Frequency distribution of HQ nitrate concentration for four groups.

Acknowledgment

None.

Conflict of Interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.dib.2020.105374](https://doi.org/10.1016/j.dib.2020.105374).

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